

Stanford Quad Study Instructions and Script

Roles:

Researcher A: Reads script to participants and measures the arm length and eye distance measurements.

Researcher B: Records measurements and raw data into the data sheet

Presentation Order:

There are four different presentation orders of the objects are used for each version of the study.

Version 1

Order 1: Lamp bulb 1(a), Lamp bulb 2(a), Lamp bulb 3(a), Lamp bulb 4(a) , Lamp post 1, Lamp post 2, Lamp post 3, Lamp post 4

Order 2: Lamp bulb 4(a), Lamp Lamp bulb 3(a), Lamp bulb 2(a), Lamp bulb 1(a), Lamp post 4, Lamp post 3, Lamp post 2, Lamp post 1

Order 3: Lamp post 1, Lamp post 2, Lamp post 3, Lamp post 4, Lamp bulb 1(a), Lamp bulb 2(a), Lamp bulb 3(a), Lamp bulb 4(a)

Order 4: Lamp post 4, Lamp post 3, Lamp post 2, Lamp post 1, Lamp bulb 4(a), Lamp bulb 3(a), Lamp bulb 2(a), Lamp bulb 1(a)

Version 2

Order 5: Lamp bulb 1a, Lamp bulb 2a, Lamp bulb 3a, Lamp 4a, Moon ball 1, Moon ball 2, Moon ball 3, Moon ball 4

Order 6: Lamp bulb 4a, Lamp bulb 3a, Lamp bulb 2a, Lamp bulb 1a, Moon ball 4, Moon ball 3, Moon ball 2, Moon ball 1

Order 7: Moon ball 1, Moon ball 2, Moon ball 3, Moon ball 4, Lamp bulb 1a, Lamp bulb 2a, Lamp bulb 3a, Lamp bulb 4a

Order 8: Moon ball 4, Moon ball 3, Moon ball 2, Moon ball 1, Lamp bulb 4a, Lamp bulb 3a, Lamp bulb 2a, Lamp bulb 1a

Version 3

Order 1: Blue ball, Lamp bulb 5, Lamp bulb 6, Lamp bulb 7, Lamp post 10, Lamp post 7, Lamp post 8, Lamp post 9

Order 2: Lamp bulb 7, Lamp Lamp bulb 6, Lamp bulb 5, Blue ball, Lamp post 9, Lamp post 8, Lamp post 7, Lamp post 10

Order 3: Lamp post 10, Lamp post 7, Lamp post 8, Lamp post 9, Blue ball, Lamp bulb 5, Lamp bulb 6, Lamp bulb 7

Order 4: Lamp post 9, Lamp post 8, Lamp post 7, Lamp post 10, Lamp bulb 7, Lamp bulb 6, Lamp bulb 5, Blue ball

Once Participant Arrives:

They might be looking lost and/or wandering around the quad. Once you find them, have them fill out the consent form (confirm you use the correct form for paid pool/ class credit participants) and ensure they sign the proper page (second to last page). The last page should be signed by the researcher obtaining the consent. Then, have them fill out the demographics information on the spreadsheet. Make sure all previous rows are hidden, and inform them that their demographic information is both confidential and optional. Have them seated at the table. Participants cannot move the location of the table but can adjust their seating position.

Introduction/Orienting the subject on the lamp locations

Bold items are notes about the script and are not to be read aloud.

Italicized items are instructions/physical actions to complete.

Normal text is read aloud.

Script: Thank you for signing up to participate in our study. This is a study about perception and how you perceive the size of different objects across the quad. We will be using the lamps scattered across the quad, as well as the inflated blue ball with spikes on it off to your right. First I'm going to point out the different lamps, and then I will walk over to a lamp to indicate where your size measurements should begin. First, you have lamp 5. Lamp 5 is the lamp to the left of the inflated blue spiky ball. If you look to the right of the blue spiky ball, at the end of that planter is lamp 10. Lamp 6 is to your left. Do you see the tall palm tree on this circular planter, lamp 6 is to the right of this circular planter. Now if you go down one planter, lamp 7 is the lamp to the right of the two tall palm trees. If you go down one more circular planter, lamp 8 is underneath the purple leaves in your vision.

(If still confused, note how lamp 6 and 7 are on the same vertical axis and if you were to draw a straight line between them, that lamp 8 would be on that same line in their sight)

Do you see the two lamps on the circular planter one down from lamp 8? Lamp 9 is the lamp of the left out of those two. Now that you have the lamp positions, I'm going to walk up to Lamp 5 and show you what to consider the lamp bulb and post.

Researcher A: Walk over to Lamp 5 and explain what two parts of the lamp they will be viewing. Most people understand that the white circles represent the full lamp bulb, but people do not as easily understand where on the lamp post they should be measuring to (the circular base at the bottom, excluding the concrete base). This can cause issues if participants are incorrectly judging the size of the lamp. Thus, physically pointing out the two landmarks ensures everyone is perceiving the same thing. Walk back once you explain this.

Researcher B: Place disk/stick tray on the table in front of them, depending on presentation order.

Experiment 3: Perceptual Matching

Script: Now we may begin. This is a matching experiment. We are curious to test your perception of the size of different objects across the quad. For this first task, please keep your upper body and head as stable as you can while looking at the probes in front of you. Now, look at (first object of presentation order). While keeping your hands at your side, please verbally indicate which probe best matches the

size of (first object of presentation order). Remember to keep your head in a consistent viewing position, we need to take a measurement from the top of your nosebridge to the table in a moment.

Researcher B: Record number once they verbally indicate the probe

Continue to sit in your current position, do not move. I now need to measure the distance from the top of your nose bridge to the table.

Researcher A: Take measurement from the top of the nose bridge to the probe they indicate. Verbally say this measurement aloud

Researcher B: Record distance measurement

Now, look at (second object in presentation order) and choose the probe that best matches the size of (second object in presentation order) from where you are sitting.

Repeat for each object in presentation order.

Experiment 4: Adjusted Matching

Researcher A: Place a set of tweezers onto the tray

“Please look back to (first object in presentation order). We would now like you to measure the size of (first object in presentation order) in your visual field. To do this, please pick up the probe you think matches the size of (first object) and hold it against the object with your arms fully extended. If for some reason the first one you pick up does not match, continue to test different probes until you find a match. You may use the tweezers to help you pick up the probe. When you get the probe to perfectly match (first object in presentation order), please continue to hold it at the point where it is matching and verbally indicate to us that you’ve found a match, and we will take a measurement from the top of your nose bridge to the probe while the probe is in the air.

Researcher A: Take the nose bridge to probe measurement and say this measurement aloud

Researcher B: Visually inspect to see which probe they are holding. Record the probe chosen and distance measurement.

Repeat for each object in presentation order

Task 3: Disparity

Have you ever looked at an object with one eye closed and noticed how it seems to jump across your vision when you switch to your other eye? In this task, we want to measure the distance that each object we just measured in the first task “jumps” across your visual field when you switch from one eye to another. You will be using this caliper to do this. You will hold the caliper with your arms fully extended up to each object to measure the distance that the object jumps across your vision. I will be using this

probe (use stick “m” if starting with posts, use disk #10 if starting with bulbs) to demonstrate how to do this measurement.

So, you will hold the caliper with your arm completely extended. Hold your arm steady in the same position throughout the measurement. Now, close your left eye and keep only your right eye open. When you align the right edge of caliper with right edge of the object in your vision, it should look like this..

Researcher A: Place the probe against the right edge of the caliper

Then, once you have aligned the right edge of the caliper with the right edge of the object, then switch eyes. Everything may slightly shift in your vision. This is okay, please do not move your arms or head because your right eye is currently holding onto where you aligned the right side of the object is aligned. Now with your left yet open, align the left edge of caliper with left edge of object. It should look like this

Researcher A: Demonstrate with the probe where it should align

Once you find the left alignment, shuffle between only your right eye and left eye being open a few times to make sure that the caliper is accurate and aligned. Once you find that alignment, verbally tell us that you have found the correct alignment. Then, while you hold your arms in place, we will take a measurement of how far the caliper reading is from the top of your nose bridge.

Researcher A: Once the participant indicates the caliper is aligned with both sides, begin to take the nose bridge to caliper measurement.

Researcher B: Once the participant indicates the caliper is aligned with both sides, you must record the number the caliper is reading. Note, the caliper is in mm, thus you must convert the measurement into centimeters. Record the arm distance measurement as well.

We will repeat this process for every object we observed in the first task.

Complete the first round

We will now repeat this measurement process once more for all of the objects.

Complete the second round

Conclude experiment.

If they are a paid participant, have them fill out the payment receipt. If there is cash in the envelope and they are a U.S. citizen, you can give them \$30 cash. if they have venmo, they can also be venmo'd. If not, please make a note on the receipt that they need a gift card sent to them.